DANISH RAZA

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Objective

To seek recognition as a Distinguished Young Alumnus by showcasing sustained academic excellence, professional impact, and international engagement in GIS, remote sensing, and data-driven decision-making. Accomplish global reputation as Alumni of COMSATS University Islamabad through continued leadership and contributions.

: www.linkedin.com/in/danish-raza-784a2b12b
: https://www.researchgate.net/profile/Danish-Raza-5?ev=hdr_xprf
: https://scholar.google.com/citations?user=rwunyTYAAAAJ&hl=en

Profile

Dr. Danish Raza is an accomplished researcher and emerging academic leader whose work exemplifies innovation, impact, and international excellence in GIS, remote sensing, and machine learning based geospatial analysis. He earned his **Ph.D.** in **Cartography and Geographic Information Engineering** from **Wuhan University, China (June 2025)**, where his research aligned with the United Nations Sustainable Development Goal (Zero Hunger) introduced advanced remote sensing methodologies to address critical food sustainability challenges. With strong **multidimensional experience** in geospatial technologies, **Dr. Raza** has contributed to **national-level projects** involving **commercial satellite data.**

Currently, he is serving as an **Individual Consultant (GIS)** at The Urban Unit, Lahore, Pak, supporting **national level planning** and **policy initiatives** through advanced geospatial analysis. He is also **Visiting Faculty member** at the **CGIS, University of the Punjab**. He has also been **selected** for a **prestigious Postdoctoral Fellowship** at **King Fahd University of Petroleum & Minerals (KFUPM), Saudi Arabia**, reflecting his growing international academic recognition. He has also **published** 27 **scientific research papers** and presented his work at worldwide forums. He has engaged in several **trainings and conferences** in **China, Hong Kong SAR and Pakistan** including platforms of the **United Nations and NASA**. His core responsibilities are GIS base data development and cartography for efficient resource management and decision making.

Main Highlights

1. **Ph.D. Cartography and Geographic Information Engineering** (2020-2025)
   State Key Laboratory of Information Engineering in Surveying Mapping and Remote Sensing (LIESMARS), Wuhan University, China **(QS World Ranking: 186)**
2. Published in six years - **27 Research Articles** including in Ph.D. 6 as 1st Author **(Total Citations: 663)**
3. Crops Spectral Library – Developed **Crops Spectral Library** (CROPSPECPK) with Spectroscopy
4. Conferences and Seminars – **6** International Events **(China, United Nation, Hong Kong)**
5. Awards Received in Ph.D. **– 4 Awards**
6. **6+** Years of Relevant Experience – **Individual Consultant – GIS** at the Urban Unit - **Visiting Faculty** at University of the Punjab, Lahore, Pakistan
7. **Selected** as **Post-Doctoral Fellow** at **King Fahd University of Petroleum and Minerals, Saudi Arabia (QS World Ranking – 67)**
8. Summer Schools – **3** International Summer Schools
9. Trainings and Workshops – **44 Certifications** including **China, United Nations, NASA and Pakistan**

Education

Ph.D. Cartography and GIE (2020-2025)
Wuhan University, China

M.S. Remote Sensing & GIS (18 Years) (2016-2018)
COMSATS University, Islamabad, Pakistan

M.Sc. in Geographical Information System (16 Years) (2013-2015)
University of the Punjab, Lahore, Pakistan

Graduation (Major Subject: Computer Studies) (2011-2013)
University of the Punjab, Lahore, Pakistan

Publications

Total Citations: 663

1. **Raza D**, Shu, H., Ehsan, M., *Fan, H., et al. (2025). Evaluation of agriculture land transformations with socio-economic influences on wheat demand and supply for food sustainability. Cogent Food & Agriculture, 11(1), 1-33. https://doi.org/10.1080/23311932.2024.2448597 (**ISI-indexed Q2 Journal, Impact Factor 2.3**)
2. ***Raza D**, Shu, H., Mirza, S., Aslam, H., et al. (2025). Multicriteria Evaluation of Cropland Suitability using Multisource Datasets of Satellite Remote Sensing and Ground Observation. Physics and Chemistry of the Earth, Parts A/B/C. https://doi.org/10.1016/j.pce.2025.103978. (**ISI-indexed Q1 Journal, Impact Factor 4.1**)
3. Ehsan, M., Chen, R., Shafi, A., **Raza D** et al. (2025). An integrated study to detect geothermal anomalies based on Landsat 8, Geological, and Geophysical data sets in Southeast Pakistan. Geothermics, 131, 103399. https://doi.org/10.1016/j.geothermics.2025.103399 (**ISI-indexed Q1 Journal, Impact Factor 3.9**)
4. P. Shu., Aslam RW., Naz I,....., **Raza D** et al. (2025). Deep Learning-Based Super-Resolution of Remote Sensing Images for Enhanced Groundwater Quality Assessment and Environmental Monitoring in Urban Areas. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, https://doi.org/10.1109/JSTARS.2025.3548010 (**ISI-indexed Q1 Journal, Impact Factor 5.3**)
5. Bin, L., Aslam, R. W., Naz, I., …., **Raza D**., Zulqarnain R.M., & Said, Y. (2025). Multisensor Remote Sensing and Advanced Image Processing for Integrated Assessment of Geological Structure and Environmental Dynamics," in IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, vol. 18, pp. 16844-16857. https://doi.org/10.1109/JSTARS.2025.3583853 **Q1, IF 5.3**
6. Duan, X., Aslam, R. W., Naqvi, S. A. A,…...,**Raza D**., Zulqarnain R.M., & Said, Y. (2025). Multi-Index Assessment and Machine Learning Integration for Drought Monitoring Using Google Earth Engine," in IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, vol. 18, pp. 16497-16516, https://doi.org/10.1109/JSTARS.2025.3580652 **Q1, IF 5.3**
7. Shabbir, H., Ehsan, M., **Raza D**., Abdelrahman, K., & Kahal, A. Y. (2025). Shallow mass movement susceptibility mapping through remote sensing using AHP, WLC, and soil moisture index analysis. Advances in Space Research. 76(6). 3305-3323. https://doi.org/10.1016/j.asr.2025.07.002 (**ISI-indexed Q1 Journal, Impact Factor 2.8**)
8. Aeman, H., Shu, H.,…, **Raza D**. (2025). Spatial–temporal dynamics of evapotranspiration and its relationship with drivers of wetlands degradation. Theoretical and Applied Climatology. 156, 356. https://doi.org/10.1007/s00704-025-05568-y (**ISI-indexed Q3 Journal, Impact Factor 2.7**)
9. ***Raza D**, Shu, H., Nazeer, M., et al. (2024). Improved method for cropland extraction of seasonal crops from multi-sensor satellite data. International Journal of Remote Sensing, 45(18), 6249-6284. https://doi.org/10.1080/01431161.2024.2388864 (**ISI-indexed Q2 Journal, Impact Factor 2.6**)

10. ***Raza D**, Khushi M., Shu, H. et al. (2024). CA-ANN based LULC prediction and influence assessment on LST-NDVI using multi-temporal satellite images. Environmental Earth Sciences 83:144, 1-20. https://doi.org/10.1007/s12665-024-11467-8 (**ISI-indexed Q2 Journal, Impact Factor 2.8**)
11. Wang N, Naz I, Aslam RW, Quddoos A, Soufan, W, **Raza D**, Ishaq T, Ahmed B. (2024). Spatio-temporal dynamics of rangeland transformation using machine learning algorithms and Remote Sensing data, Rangeland Ecology & Management,94,106-118. https://doi.org/10.1016/j.rama.2024.02.008 (**ISI-indexed Q2 Journal, Impact Factor 2.4**)
12. Zhao Z, Islam F, Waseem LA, Tariq A, ..., Aslam RW, **Raza D,** Hatamleh, WA. (2024) Comparison of Three Machine Learning Algorithms Using Google Earth Engine for Land Use Land Cover Classification. Rangeland Ecology & Management. 1 (92):129-37. https://doi.org/10.1016/j.rama.2023.10.007 (**ISI-indexed Q2 Journal, Impact Factor 2.4**)
13. ***Raza, D**., Shu, H. et al. (2022). Comparative Geospatial Approach for Agricultural Crops Identification in Inter-Fluvial Plain – A Case Study of Sahiwal District, Pakistan. Pakistan Journal of Agricultural Sciences. 59(4): 567-578. https://doi.org/10.21162/PAKJAS/22.127 (**ISI-indexed Q3 Journal, Impact Factor 0.8**)
14. Khushi M, ***Raza D**, Kamran M, Nazeer M, Ahmad A, Khan SU, Younas I. (2023). Delineation of groundwater potential zones with Analytic Hierarchy Process based geospatial modelling approach in metropolitan expanse. Desalination and Water Treatment. 315, 399-412. https://doi.org/10.5004/dwt.2023.30155 (**ISI-indexed Q4 Journal, Impact Factor 1.0**)
15. Aslam, R. W., Shu, H., Javid, K., Pervaiz, S., Mustafa, F., **Raza D**., ... & Hatamleh, W. A. (2023). Wetland Identification through Remote Sensing: Insights into Wetness, Greenness, Turbidity, Temperature, and Changing Landscapes. Big Data Research. 35, 100416. https://doi.org/10.1016/j.bdr.2023.100416 (**ISI-indexed Q1 Journal, Impact Factor 4.2**)
16. Ghalib A, Mirza S, ***Raza D**, Rafi A, Ali F. (2023). Valuation of Socio-Economic indicators for progressing urban sustainability under distinctive rubrics. Wiley Library. Transactions in GIS. 00. 1-20. https://doi.org/10.1111/tgis.13047 (**ISI-indexed Q1 Journal, Impact Factor 2.3**)
17. Sohail, M. T., Manzoor, Z., Ehsan, M., …. **Raza, D**., Usman, U. and Akbar, S. (2023). Impacts of urbanization, LULC, LST, and NDVI changes on the static water table with possible solutions and water policy discussions: A case from Islamabad, Pakistan. Frontiers in Environmental Science, 11, 140. https://doi.org/10.3389/fenvs.2023.1018500 (**ISI-indexed Q2 Journal, Impact Factor 3.7**)
18. ***Raza D**., Mirza, S., Shu, H. et al. (2023). Geospatial Approach for Petrol Pumps Valuation with Urban Prediction Modelling by Cellular Automata in Creeds of Metropolitan Expanse. ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences, 10, 59-67. https://doi.org/10.5194/isprs-annals-X-5-W1-2023-59-2023
19. Amjad D, Mirza S, ***Raza D**, Sarwar F, and Kausar S. (2023). A Statistical Modeling for spatial-temporal variability analysis of solar energy with respect to the climate in the Punjab Region: A Statistical Modeling for spatial-temporal variability analysis of solar energy. Bahria University Research Journal of Earth Sciences, 7: 1 – 10. https://burjes.bahria.edu.pk/index.php/ojs/article/view/48
20. Mirza S, Butt HJ, Khalid I, **Raza D**, Akmal F, Khan SU. (2022). Spatial Site Selection for Industries using Decision Rules- A case study of Sargodha Division. Fresenius Environmental Bulletin. 31(6): 5409-4418. https://www.prt-parlar.de/download_list/?c=FEB_2022
21. Mehmood A, Qadir A, Ehsan M, Ali A, **Raza D**, Aziz H. (2021). Hydrogeological Studies and Evaluation of Surface and Ground Water Quality of Khyber Pakhtunkhwa, Pakistan. Desalination and Water Treatment. 244: 41-54. https://doi.org/10.5004/dwt.2021.27913 (**ISI-indexed Q4 Journal, Impact Factor 1.0**)
22. Ullah H, Aslam RW, Ahmed SR,…..., **Raza D**. (2021). Geo-spatial analysis and modelling of water borne diseases in District Multan. J. Bio. & Env. Sci. 18(9), p. 1-15. https://www.innspub.net/wp-content/uploads/2022/05/JBES-V19-No3-p23-35.pdf

23. Khan SU, ***Raza D**, Saeed U, …..., Ali M, Ahmad SR. (2020). Evaluation of foliage spread conversion into industrial land under the significance of urbanization and Environment. Fresenius Environmental Bulletin. 29(6): 4612-4623. https://www.prt-parlar.de/download_list/?c=FEB_2020
24. J. Iqbal, M. Ali, A. Ali, **Raza D**, F. Bashir, F. Ali, S. Hussain, Z. Afzal. (2020). Investigation of cryosphere dynamics variations in the upper Indus basin using remote sensing and GIS. The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, Volume XLIV-3/W1 2020. https://doi.org/10.5194/isprs-archives-XLIV-3-W1-2020-59-2020
25. **Raza D**, Ehsan M, Khan SU, Raza A, Saeed U, Akmal F. (2020). Dire situation of COVID 19 in Pakistan with its impact on aerosol optical depth. BURJES. 5(1): 1-4. https://drive.google.com/file/d/1rv42XWPVWxkwy-fypQ2xfmRm30V-w1oJ/view
26. ***Raza D**, Karim RB, Nasir A, Khan SU, Zubair MH, Amir R. (2019). Satellite based surveillance of LULC with deliberation on urban land surface temperature and precipitation pattern changes of Karachi, Pakistan. J Geogr Nat Disast. 9: 237. doi: 10.4172/2167-0587.1000237
27. ***Raza D**, Kidwai AA. (2018). Geospatial analysis of Indus river meandering and flow pattern from Chachran to Guddu barrage, Pakistan. RADS J. Biol. Res. Appl. Sci. 9(2): 1-8. https://doi.org/10.37962/jbas.v9i2.155

Conferences & Seminars

1. **Raza D,** Shu H, Fan H. (2025). Impact Assessment of Agricultural Land Conversion Using Socio-Economic and Remote Sensing Data for Sustainable Food System. at **SPIE conference on Sensing for Agriculture and Food Quality and Safety XVII** under SPIE Defense + Commercial Sensing (2025).
2. **Raza D,** Shu H, Fan H. (2025). Agricultural Land Transition and its Influencing Impact on Food Sustainability. at **United Nations International Conference** under United Nations Office for Outer Space Affairs and UN-SPIDER (20-22 January, 2025) at Beijing, **China**.
3. **Raza D**, Mirza, S., Shu, H. et al. (2023). Geospatial Approach for Petrol Pumps Valuation with Urban Prediction Modelling by Cellular Automata in Creeds of Metropolitan Expanse. **ISPRS International Conference on Geomatics Education**. (10-12 May, 2023) The Hong Kong Polytechnic University, HK. – **Hong Kong, SAR China.**
4. **Raza D**, Shu, H. (2024). Remote Sensing for Seasonal Agricultural Crops Assessment with Satellite and In-Situ Data. **International Student Aerospace Academic Seminar**. Host: China Scholarship Council, Organizer: Beihang University, Hangzhou, China July, 2024. (Excellent Poster Award) - **China**
5. **Raza D**, Shu H. (2023). Cropland Identification and Temporal Behavior Assessment using Satellite Data. OPENDAY-LIESMARS, at Wuhan University, China. March, 2023 – **China**.
6. **Raza D**, Shu H. (2023). Agricultural Crop Behavior Assessment with Optical and Thermal Satellite Data. OPENDAY-LIESMARS, at Wuhan University, China. October, 2023 – **China**.

Other Publications

1. Map Book of the Punjab. (https://urbanunit.gov.pk/enewsletters/urbanatlas/index.html#p=1)
2. Development of Mauza Boundaries Dashboard of the Punjab (2023) under Space Technology Applications in Socio-Economic Development Progress. Final Mauza Dashboard (urbanunit.gov.pk).

Summer School International

1. Attended Luojia Global Summer School on **"Social Sensing and Big Data Computing for Disaster Management"** from 23- 29 June, 2024. Arranged by **International Office & LIESMARS, Wuhan University, China**.

2. Attended International Geo-Informatics Summer School on **"Smart Earth"** from 21- 27 August, 2022. Arranged by **LIESMARS, Wuhan University, China**.
3. Attended International Geo-Informatics Summer School on **"Machine Learning"** from 23- 31 August, 2021. Arranged by **LIESMARS, Wuhan University, China**.

Trainings & Workshops

1. Attended the **United Nations International Training** on "**New Satellite Resources for Emergency Response: Disasters Charter, Charter Mapper, and Other Commercial Solutions**" in Beijing, China (17-18, Jan 2025). Organized by **United Nations Office for Outer Space Affairs** and **UN-SPIDER**, Beijing, China
2. Attended training workshop title **as "Geospatial Technology and Industry".** On 23 - 26 May, 2023. at **Hi-Target Satellite Navigation Technology** Co., Ltd, Guangzhou, China.
3. Attended training title as **"Remote Sensing of Night-time Light and its Application in Sustainable Development"**. On 13 - 14 October, 2023. Arranged by **LIESMARS, Wuhan University**, Wuhan, Hubei, China.
4. Attended training title as **"Calculating Spectral Indices for Land and Aquatic Applications Using QGIS"**. Feb, 2025. Arranged by **NASA's Applied Remote Sensing Training** Program.
5. Attended training title as **"Building Climate Risks Assessments from Local Vulnerability and Exposure"**. On 19 - 21 September, 2023. Arranged by **NASA's Applied Remote Sensing Training** Program.
6. Attended training title as **"Transforming Earth Observation (EO) Data into Building Climate Risk Assessments from Local Vulnerability and Exposure"**. On 03, 05 & 10 October, 2023. Arranged by **NASA's Applied Remote Sensing Training** Program.
7. Attended training workshop title as **"International Training Workshop on Earth Observations for Sustainable Development in Developing Countries"** from 26 to 28 July, 2022. Jointly Organized by **ICIMOD**, **NRSCC** and **AOGEO**.
8. Attended training workshop title as **"International Training Workshop on Resource & Environment Scientific Data Sharing and Disaster Risk Reduction Knowledge Service along the Belt and Road"** from 20 to 24 November, 2022 under Auspices of **UNESCO (IKCEST)**.
9. Attended training title as **"Monitoring and Modeling Floods using Earth Observations"**. On 14 & 21 September, 2022. Arranged by **NASA's Applied Remote Sensing Training** Program.
10. Attended course title as **"Remote Sensing of Night-Time Light and its Application in Sustainable Development"** on 27th Nov - 2th Dec, 2022. Hosted by Deqing Association for Geospatial Information Professionals, China (DQGIP) under The **Belt and Road International Geospatial Information Training Center BRGTC China**.
11. Attended training title as **"Earth Observations Toolkit for Sustainable Cities and Human Settlements"** On Feb, 2022. Arranged by **NASA's Applied Remote Sensing Training** Program.
12. Attended course title as **"Hydrogeodesy"** on 28th Nov - 22th Dec, 2022. Hosted by Deqing Association for Geospatial Information Professionals, China (DQGIP) under **The Belt and Road International Geospatial Information Training Center BRGTC China**.
13. Attended training title as **"Satellite Remote Sensing for Measuring Urban Heat Islands and Constructing Heat Vulnerability Indices"**. On 2,4,9 & 11 August, 2022. Arranged by **NASA's Applied Remote Sensing Training** Program.

14. Attended training title as **"Applications of Remote Sensing-Based Evapotranspiration Data Products for Agricultural and Water Resource Management"**. On 1, 8 & 15 June, 2022. Arranged by **NASA's Applied Remote Sensing Training** Program.
15. Participated in training entitled as **"Precision Mapping using Advanced Survey Techniques of UAV, Lidar and GNSS-RTK"** from 03 to 07 October 2022. Arranged by **SUPARCO**, Pakistan.
16. Attended course title as **"Agricultural Remote Sensing"** on 4th-5th July, 2022. Hosted by Association of Science and Technology Professionals, Deqing Geospatial Information Town PRC (DQGAST) under The **Belt and Road International Geospatial Information Training Center BRGTC China**.
17. Attended training title as **"Mapping Crops and their Biophysical Characteristics with Polarimetric Synthetic Aperture Radar and Optical Remote Sensing".** On 12,19, 26 April & 3 May, 2022. Arranged by **NASA's Applied Remote Sensing Training** Program.
18. Attended training title as **"Using the UN Biodiversity Lab to Monitor the Pulse of the Planet".** On 14,21,27,28 April & 4 May, 2022. Arranged by **NASA's Applied Remote Sensing Training** Program.
19. Attended training title as **"Using Google Earth Engine for Land Monitoring Applications".** On 16,23 & 30 June, 2021. Arranged by **NASA's Applied Remote Sensing Training** Program.
20. Participated in training entitled as **"Satellite based Crop Monitoring, Yield Estimation/ Forecasting and Land Cover Mapping"** from 29 Nov to 03 December 2021. Arranged by **SUPARCO, Pakistan**.
21. Participated in training entitled as **"Drone Flight"** on 27-29 June, 2022. Hosted by The **Urban Unit**, **Pakistan**.
22. Attended training entitled as **"UAV LIDAR Data Acquisition & Processing for Urban Areas"** on November 17, 2021. Arranged by **The Urban Unit Lahore, Pakistan**.
23. Attended training title as **"Introduction and Access to Global Air Quality Forecasting Data and Tools".** On 23,28 & 30 Sep, 2021. Arranged by **NASA's Applied Remote Sensing Training** Program.
24. Attended training title as **"Monitoring Coastal and Estuarine Water Quality - Transitioning from MODIS to VIIRS".** On 14,16 & 21 Sep, 2021. Arranged by **NASA's Applied Remote Sensing Training** Program.
25. Attended training title as **"Introduction to Population Grids and their Integration with Remote Sensing Data for Sustainable Development and Disaster Management"**. On March 30 and April 6, 2021. Arranged by **NASA's Applied Remote Sensing Training** Program.
26. Attended a session title as **"Artificial Intelligence and Machine Learning for Environmental Research and Application"** on 03 Sep 2020. Hosted by **American Meteorological Society** Webinars.
27. Attended a Training session title as **"MODIS to VIIRS Transition for Air Quality Applications"** on 22 Oct 2020. Arranged by **NASA's Applied Remote Sensing Training** Program
28. Attended the Seminar title as **"The Role and Position of Cartography and GIS in Geographic Information Science"**. Arranged by LIESMARS, **Wuhan University** on 14 Oct 2020. Speaker: Professor Wolfgang Kainz
29. Attended Seminar title as **"The Challenges of Quantifying Global Warming".** On 09 Dec, 2020. Arranged by **LIESMARS**, Wuhan University China.
30. Attended the Seminar title as **"Geomatics in Ocean Science"**. Arranged by **LIESMARS,** Wuhan University on 11 Nov 2020. Speaker: Armin Gruen
31. Attended multiple training sessions on topic **"Satellite Remote Sensing for Urban Heat Island"** on 10, 17 and 24 Nov 2020. Arranged by **NASA's Applied Remote Sensing Training** Program.

32. Attended a Training session title as **"Introduction to NASA's "Black Marble" Night Lights Data"** on 03 Dec 2020. Arranged by **NASA's Applied Remote Sensing Training** Program.
33. Attended Training sessions on **"Hyperspectral Data for Land and Coastal Systems"** on Dated Jan 19, 26 and Feb 02, 2021. Arranged by **NASA's Applied Remote Sensing Training** Program.
34. Attended Training sessions on **"Species Distribution Modeling with Remote Sensing"** on Dated 12, 17 & 19 Aug 2021. Arranged by **NASA's Applied Remote Sensing Training** Program.
35. Attended Training sessions on **"Connecting Citizen Science with Remote Sensing"** on Dated Jan 24, 26 and 31 2023. Arranged by **NASA's Applied Remote Sensing Training** Program.
36. Participated in training entitled as **"Image Classification, OBIA and Google Earth Engine"** on 14 to 17 December 2020. Arranged by The **Urban Unit, Pakistan**.
37. Attended Training sessions on **"Use of Solar Induced Fluorescence and LIDAR to Assess Vegetation Change and Vulnerability"**. 16-25 March, 2021. Arranged by **NASA's Applied Remote Sensing Training** Program.
38. Attended Training sessions on **"Monitoring Coastal and Estuarine Water Quality Using Remote Sensing and In SITU Data"**. 30 Nov & 07 Dec, 2021. Arranged by **NASA's Applied Remote Sensing Training** Program.
39. Attended Training sessions on **"Satellite Observations and Tools for Fire Risk, Detection and Analysis".** 11- 27 May, 2021. Arranged by **NASA's Applied Remote Sensing Training** Program.
40. Participated in training entitled as **"Drone Imaging and Advanced Image Processing"** on 02-03 Mar, 2021. Hosted by The **Urban Unit, Pakistan**.
41. Three Days Training Workshop on **"Solar Pumps and High Efficient Irrigation System Design and Management".** Organized by Directorate of short courses, **PCC Unit Pak-China** and **CAEWRI**" National Agricultural Research Centre (**NARC**), Islamabad, Pakistan
42. Certificate of Participant in International workshop on **"Precision Agriculture and Forestry".** Lahore University of Management Sciences **(**LUMS**), Pakistan**
43. Certificate of Symposium on **"Gravitational waves Astrophysics – Sources, detection and Future Implication" Institute of Space Technology, Islamabad, Pakistan**
44. Certificate in **"Information System"**
    Virtual University of Pakistan

Experiences

Visiting Faculty - Machine Learning for Spatial Data, Spring, 2026
Centre for Geographical Information System
University of the Punjab, Lahore, Pakistan

- Taught Machine Learning techniques for spatial and geospatial datasets, focusing on GIS and Remote Sensing applications
- Covered supervised and unsupervised learning methods
- Applied ML models for classification**,** environmental monitoring**,** and urban analysis
- Conducted hands-on practical sessions using Python**,** Google Earth Engine**,** and GIS platforms (ArcGIS Pro/QGIS)
- Guided students in model training, validation, accuracy assessment, and interpretation of spatial ML results

Professional Experience **6 Years +**

- **Individual GIS Consultant**
  The Urban Unit, Lahore, Pakistan (P&D Board) **Currently Working**
  - Supervised a 50+ member GIS professional team for national-level geospatial projects.
  - Providing specialized GIS and remote sensing consultancy to support spatial analysis, policy-oriented studies, and decision-making for government led projects.
  - Supporting advanced geospatial analysis, data validation, and cartographic outputs for urban planning and resource management initiatives.
- **As "Research Associate Remote Sensing/GIS" at "The Urban Unit, Lahore, Pakistan" (P & D). (From April, 2017- December, 2022)**
  - Ability to handle and process the satellite data and remote-sensing based tasks on Province and Country level as hand on expertise to achieve deadline of assigned tasks in diverse known national level projects such as UIPT, PSS, NTRC, SDPI, Space Tech. etc., similarly in other short assignments.
  - Remote Sensing for earth features extraction. Geographic Object based Image Analysis
  - Ability to recognize and extraction of Water cover, Land features, Forest cover as well Urban tree cover with RS & GIS methodologies
  - Satellite image Pre-processing to prepare the Raster Data. Maintain Satellite's Metadata.
  - Image correction and Rectification, Image difference analysis, Satellite Image Classification, features extraction with the help of Geographic object-based image analysis
  - Validation of data, Accuracy Assessment, processing and automation with optimized methods
  - Conversant to handle bulks of satellite imagery also hand on experience for processing of high and low resolution Satellites i.e. WorlView 1, WorlView 2, WorldView 3, QuickBird, GeoEye 1, Ikonos, Landsat, Sentinel etc. and much more AirBus and Digital Globe satellites as well as Aerial imagery
  - Quality Assurance of the downloaded seen, generate of report according to Nadir angle, Cloud cover, Blending, Stretching and area missing contrast.
  - Agricultural crop identification through remote sensing; GPS based survey for crop signature collection
  - Any-other tasks assigned by Team Lead.

- **As "GIS Analyst" at Mott MacDonald Pakistan (MMP). (Feb-2017 - April-2017)**
  - Monitored and evaluate the vector data of Water supply, Sewerage, drainage network.
  - Maintained the accuracy of data by the meeting with responsible authorities of WASA Lahore.
  - Geo-Referenced the Land Record of WASA Assert and Private Housing society's maps.
  - Surveyed for boundary demarcation of the unidentified place (Ground truthing)
  - Analyzed complete Dataset and Export Maps with WASA Specified Symbols.

Awards & Achievements

1. Received Outstanding International Doctoral Student Award (1st Prize) – 2024 in All Academic Achievements by Wuhan University, China
2. Received Outstanding International Doctoral Student Award (2nd Prize) – 2023 in All Academic Achievements by Wuhan University, China

3. Received Full Scholarship for Doctoral Studies – 2020 by China Scholarship Council (CSC).
4. Excellent Poster Award at International Student Aerospace Seminar – 2024. Hosted By CSC, China

Societies and Memberships

1. Member of International Society of Photo-Optical Instrumentation Engineers (SPIE).
2. Member of IEEE Geoscience and Remote Sensing Society (IEEE-GRSS)

Personal Information

Full Name	Danish Raza
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